

Project 1 Title: LLM-Driven Risk Identification and Analysis

What's the goal?

The main idea is to **use AI (like ChatGPT or Llama-2)** to automatically **read project documents** (like requirement documents, meeting notes, etc.) and **identify possible risks** in the project.

For example:

- If the document says “the team is short-staffed,” the AI should say: “**Risk: Delay due to lack of manpower (Category: Schedule)**”

Research Questions

1. Can AI read project documents and spot risks on its own?
2. How good is AI at finding risks compared to human experts?
3. Can AI also suggest ways to handle or reduce those risks?

How will we do it?

- Use a smart AI model (like GPT-4 or Llama-2).
- Give it project-related text and ask it to **list risks** with labels like:
 - Scope risk
 - Budget risk
 - Time/schedule risk
- Use smart techniques like:
 - Giving examples in the prompt (few-shot prompting)
 - Step-by-step thinking (chain-of-thought prompting)
 - Matching known risk phrases from a database (retrieval-augmented prompting)

What data will we use?

- **Real data is hard to find**, so we can **create fake (synthetic) data**:
 - Make up short project stories or meeting notes
 - Manually mark what the risks are (this is your ground truth)
 - Use case studies or GitHub issues as realistic examples
 - Use official risk examples (like from PMBOK) to help create fake examples

What will we deliver?

1. A simple tool (like a notebook or app) where you paste project text, and the AI tells you the risks.
2. A mini “risk dataset” – project texts + risk labels.
3. Sample prompts you used with the AI.
4. A report that shows:

- How well the AI did (accuracy, precision)
 - How it compares to basic methods (like keyword search)
 - How changing the prompt affects results
5. (Optional) If the AI suggests ways to handle the risk, include that too.

Why is this useful?

Project managers spend time manually reading documents and figuring out risks. If AI can do this reliably, it saves time and improves decision-making. This project explores if that's possible and creates tools and data others can build on.

Project 2 Title: Using Generative AI to Create Project Management Documents

What's the Goal?

This project is about **building an AI tool** that can automatically write common project management (PM) documents—like **project charters**, **status reports**, or **meeting summaries**—**from just a few inputs** (like bullet points about the project).

What's the Research Question?

Can a **Generative AI** (like GPT-4 or LLaMA) create useful and clear project documents if we give it **minimal input**?

Also, **what kind of prompts (instructions)** work best to get accurate and well-written documents?

How Will You Do It?

- Use **prompt engineering**: carefully design questions or commands to give to the AI.
- Try ideas like **prompt chaining**:
 1. First, ask the AI to list headings (e.g. Goals, Scope, Timeline)
 2. Then, ask it to write each section under those headings.
- Use tools like **LangChain** to make the AI's process more structured.
- Optionally, **fine-tune a smaller AI model** if you have enough examples.

Where's the Data Coming From?

- Use **free templates** from project management books or websites.
- Make your own **sample inputs and outputs** if needed (e.g. a short project description → full project charter).
- Use **synthetic data** if real examples aren't available.

What Will You Deliver?

- A **working AI assistant** that:
 - Takes in basic project details (like goals, team, timeline)
 - Generates a full draft document (like a project charter or meeting minutes)
- Example **prompts and results** to show how it works
- A **codebase** that runs the system
- A **review or scorecard** to check how good the AI's output is (maybe done by your classmates or project advisors)

Why Is This Useful?

- It shows how AI can **save time** in routine project tasks.
- It gives hands-on experience with **prompt design**, **LLMs**, and **document generation**.
- It fits into the growing trend of AI **agents** that can read instructions, generate content, and assist in decision-making.